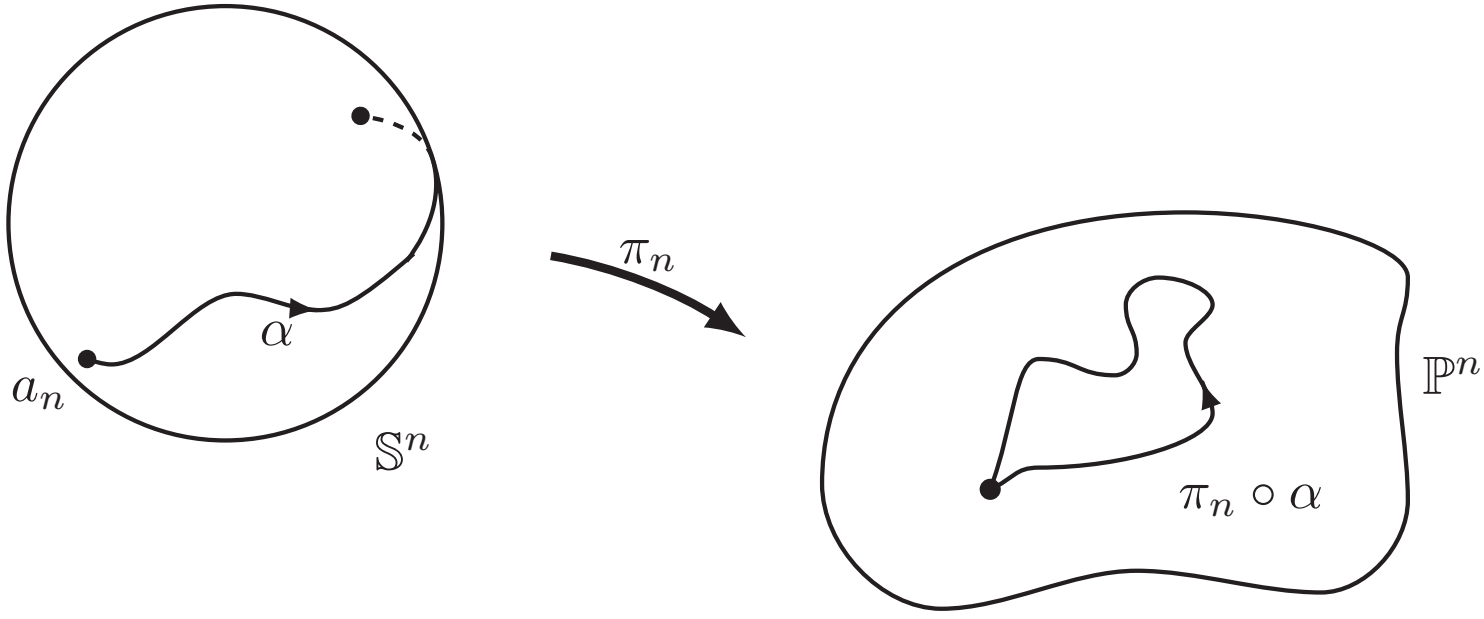

a_n
α
$\mathbb{S}^n$
π_n
$\mathbb{P}^n$
$\pi_n \circ \alpha$